

Cost Investigation – July 2026 [Synthetic]

1 View CUDOS Dashboard Demo

Here's the **CUDOS Dashboard Demo [Synthetic]** data for **July 2026** in region **ap-southeast-2 (Sydney)** — read-only view for investigation purposes :

1.1 Total Cloud Cost — July 2026

\$1,180.00 USD (Net Unblended Cost)

1.2 Service Cost Breakdown

Service	Net Unblended Cost (USD)	% of Total
AmazonEC2	\$693.30	58.8%
AmazonRDS	\$266.60	22.6%
AWSLambda	\$99.20	8.4%
AmazonS3	\$83.70	7.1%
AWSDataTransfer	\$37.20	3.2%

1.3 Top Cost Drivers (by Resource)

Service	Owner	Environment	Resource ID	Cost (USD)
AmazonEC2	Platform Team	production	i-demo-platform-01	\$558.00
AmazonRDS	Data Team	production	db-demo-rds-01	\$266.60
AmazonEC2	Demo Workloads	staging	i-demo-staging-01	\$135.30
AWSLambda	Application Team	production	function-demo-inference	\$99.20
AmazonS3	Data Team	production	bucket-demo-analytics	\$83.70
AWSDataTransfer	Unallocated	shared	data-transfer-demo	\$37.20

1.4 Net Unblended Cost — Daily Trend

The daily cost trend shows a notable step-down around July 7–8:

- **Jul 1–7:** ~\$39.50–\$39.90/day (higher spend period)
- **Jul 8–31:** ~\$37.40–\$37.70/day (stabilized lower spend)

This suggests a cost optimization event or resource change occurred around July 7–8.

Chart 1

1.5 Key Observations

- **Largest cost driver:** EC2 instance **i-demo-platform-01** (Platform Team, production) at \$558 — nearly half of total spend
- **Cost reduction event:** Daily spend dropped ~\$2.20/day (~5.5%) around July 7–8, saving approximately \$52 over the remainder of the month
- **Unallocated costs:** \$37.20 in data transfer costs remain untagged — consider tagging for better attribution

Source: CUDOS Dashboard Demo [Synthetic], Sheet 1. No additional filters were available on this dashboard — the data shown is pre-scoped to July 2026, ap-southeast-2 (Sydney).

2 Review Observed Evidence Summary

Here is the summary of the observed cost evidence based solely on the inputs you provided:

2.1 **SYNTHETIC DEMONSTRATION — Not Real CUR Billing Data**

The following summary is derived entirely from the **CUDOS Dashboard Demo [Synthetic]** and does not represent actual AWS Cost and Usage Report (CUR) billing data.

2.2 **Analysis Period & Scope**

- **Period:** July 2026
- **Region:** ap-southeast-2 (Sydney)
- **Source:** CUDOS Dashboard Demo [Synthetic]

2.3 **Top Service Evidence**

- **Amazon EC2** was the highest-cost service in the analysis period, recording a **net unblended cost of USD 693.30** in the Sydney region.

2.4 **Staging Environment — Before/After Observation**

Metric	Value
Baseline (1–7 July 2026)	USD 42.00
Measurement (8–14 July 2026)	USD 27.00
Realized Reduction	USD 15.00 (35.71%)

2.5 **Summary of Observations**

The synthetic dataset shows that within the staging environment, weekly spend dropped from USD 42.00 to USD 27.00 between the baseline and measurement windows — a **USD 15.00 reduction representing a 35.71% decrease**. Amazon EC2 dominates overall regional spend at USD 693.30 net unblended cost for July 2026.

All figures above are synthetic and provided for demonstration purposes only. No workload-changing actions are recommended based on this evidence.

3 **Review Financial Impact Analysis**

3.1 **Financial Impact Narrative**

Input Summary

Parameter	Value
Baseline Cost	SYNTHETIC USD 42.00 (1–7 July 2026)
Measurement Cost	SYNTHETIC USD 27.00 (8–14 July 2026)
Realized Reduction (Absolute)	SYNTHETIC USD 15.00
Realized Reduction (Percentage)	SYNTHETIC 35.71%
Materiality Threshold – Percentage	20%
Materiality Threshold – Absolute	USD 10.00

Verification of Provided Figures

- **Absolute reduction:** SYNTHETIC USD 42.00 – SYNTHETIC USD 27.00 = **SYNTHETIC USD 15.00**
- **Percentage reduction:** SYNTHETIC USD 15.00 ÷ SYNTHETIC USD 42.00 × 100 = **SYNTHETIC 35.71%**

Both figures are internally consistent with the provided inputs.

Materiality Assessment

Criterion	Threshold	Realized Value	Met?
Percentage reduction	20%	SYNTHETIC 35.71%	✓ Yes
Absolute USD reduction	USD 10.00	SYNTHETIC USD 15.00	✓ Yes

Determination

The reduction meets the materiality threshold.

Both required criteria are satisfied simultaneously:

1. The SYNTHETIC percentage reduction of **35.71%** exceeds the 20% percentage threshold by **15.71 percentage points**.
2. The SYNTHETIC absolute reduction of **USD 15.00** exceeds the USD 10.00 absolute threshold by **SYNTHETIC USD 5.00**.

Because the defined materiality standard requires **both** criteria to be met concurrently, and both are confirmed met, the observed week-over-week cost movement from **SYNTHETIC USD 42.00** (1–7 July 2026) to **SYNTHETIC USD 27.00** (8–14 July 2026) is classified as a **material reduction** under the stated threshold parameters.

All cost figures are labeled SYNTHETIC. No external systems were accessed. No workload-changing actions are recommended.

4 Review Final FinOps Investigation Report

 SYNTHETIC DEMONSTRATION — Not Real CUR Billing Data

COST INVESTIGATION REPORT — July 2026 [SYNTHETIC]

Report Type: FinOps Cost Investigation — Read-Only **Data Classification:** SYNTHETIC **Source Dashboard:** CUDOS Dashboard Demo [Synthetic] **Report Status:**  REVIEW REQUIRED **Generated For:** Demo Platform Team

4.1 1. OBSERVED EVIDENCE

Source: CUDOS Dashboard Demo [Synthetic] | **Period:** July 2026 | **Region:** ap-southeast-2 (Sydney)

Top Service Evidence Based on the CUDOS Dashboard Demo [Synthetic], **Amazon EC2** was identified as the highest-cost service in the ap-southeast-2 (Sydney) region during the analysis period, recording a **net unblended cost of USD 693.30** for July 2026. This positions Amazon EC2 as the dominant cost driver within scope and the primary service warranting investigation.

Staging Environment — Before/After Observation Within the staging environment subset of the Amazon EC2 spend, the following week-over-week cost movement was observed across two discrete measurement windows:

Observation Window	Weekly Cost (SYNTHETIC)
Baseline: 1–7 July 2026	USD 42.00
Measurement: 8–14 July 2026	USD 27.00
Realized Reduction (Absolute)	USD 15.00
Realized Reduction (Percentage)	35.71%

The synthetic dataset shows a clear and measurable week-over-week decline in staging environment spend. Amazon EC2 dominates the overall regional cost profile at USD 693.30 net unblended cost for the full July 2026 period. The staging environment reduction of USD 15.00 (35.71%) represents a notable movement within that broader EC2 cost envelope.

 All figures are SYNTHETIC and sourced exclusively from the CUDOS Dashboard Demo [Synthetic]. These do not represent actual AWS Cost and Usage Report (CUR) billing data.

4.2 2. FINANCIAL IMPACT

Source: Financial Impact Analysis derived from CUDOS Dashboard Demo [Synthetic] inputs.

Input Parameters

Parameter	Value
Baseline Cost	SYNTHETIC USD 42.00 (1–7 July 2026)
Measurement Cost	SYNTHETIC USD 27.00 (8–14 July 2026)
Realized Reduction — Absolute	SYNTHETIC USD 15.00
Realized Reduction — Percentage	SYNTHETIC 35.71%
Materiality Threshold — Percentage	20%
Materiality Threshold — Absolute	USD 10.00

Verification of Figures Both provided figures have been verified for internal consistency against the supplied inputs:

- **Absolute reduction:** SYNTHETIC USD 42.00 – SYNTHETIC USD 27.00 = **SYNTHETIC USD 15.00**
- **Percentage reduction:** SYNTHETIC USD 15.00 ÷ SYNTHETIC USD 42.00 × 100 = **SYNTHETIC 35.71%**

No external systems were accessed to produce or validate these figures.

Materiality Assessment

Criterion	Threshold	Realized Value (SYNTHETIC)	Threshold Met?
Percentage Reduction	20%	35.71%	 Yes — exceeds threshold by 15.71 percentage points
Absolute USD Reduction	USD 10.00	USD 15.00	 Yes — exceeds threshold by USD 5.00

Materiality Determination

 **The observed cost reduction meets the defined materiality threshold under both required criteria.**

The materiality standard requires **both** criteria to be satisfied concurrently. Both are confirmed met:

1. The SYNTHETIC percentage reduction of **35.71%** exceeds the 20% percentage threshold.
2. The SYNTHETIC absolute reduction of **USD 15.00** exceeds the USD 10.00 absolute threshold.

The week-over-week cost movement from **SYNTHETIC USD 42.00** (1–7 July 2026) to **SYNTHETIC USD 27.00** (8–14 July 2026) is therefore classified as a **material reduction** under the stated threshold parameters and warrants formal investigation and human verification.

 All cost figures are labeled SYNTHETIC. This determination is based solely on the provided inputs and does not constitute a recommendation for any workload-changing action.

4.3 3. SCOPE

Scope Parameter	Value
Region	ap-southeast-2 (Sydney)
Analysis Period	July 2026
Primary Service in Focus	Amazon EC2 (derived from top service evidence: USD 693.30 net unblended cost)
Environment Sub-Scope	Staging environment (EC2 instances within ap-southeast-2)
Measurement Windows	Baseline: 1–7 July 2026 / Measurement: 8–14 July 2026
Data Classification	SYNTHETIC
Source Dashboard	CUDOS Dashboard Demo [Synthetic]
Materiality Threshold Applied	20% AND USD 10.00 (both criteria must be met concurrently)

This investigation is scoped exclusively to the Amazon EC2 service within the ap-southeast-2 (Sydney) region for the period of July 2026, with particular focus on the staging environment cost movement observed between the two defined weekly windows. No other services, regions, or time periods are in scope for this report.

 All data within this scope is classified SYNTHETIC. No real AWS account, billing record, or production workload is represented.

4.4 4. HYPOTHESIS UNVERIFIED — NOT CONFIRMED

 **CRITICAL NOTICE:** The following hypothesis is entirely speculative and has **not been verified** by any human reviewer, automated system, or external data source. It must **not** be acted upon, referenced as fact, or used to justify any workload-changing decision. It is presented solely to structure the human verification process.

Hypothesis Statement The observed reduction in staging environment Amazon EC2 costs from **SYNTHETIC USD 42.00** (1–7 July 2026) to **SYNTHETIC USD 27.00** (8–14 July 2026) — a SYNTHETIC USD 15.00 decrease representing a SYNTHETIC 35.71% reduction — may plausibly be explained by one or more of the following **unverified** scenarios:

Hypothesis A — Scheduled Instance Rightsizing or Shutdown (UNVERIFIED) One or more EC2 instances in the staging environment within ap-southeast-2 may have been rightsized to a smaller instance type, stopped, or terminated between 7 July and 8 July 2026. A reduction of approximately 35% in weekly spend is consistent with the removal or downsizing of a subset of running instances. This has **not been confirmed** against any instance state records, AWS Config history, or CloudTrail logs.

Hypothesis B — Scheduled Scaling Policy Activation (UNVERIFIED) A scheduled Auto Scaling policy or instance scheduler rule may have taken effect on or around 8 July 2026, reducing the number of running instances during off-peak hours or non-business days within the staging environment. This has **not been confirmed** against any Auto Scaling group configuration, instance scheduler logs, or operational runbooks.

Hypothesis C — Workload Migration or Decommission (UNVERIFIED) A staging workload may have been migrated away from on-demand EC2 instances — for example, to a containerized or serverless architecture — or a staging environment may have been partially decommissioned between the two measurement windows. This has **not been confirmed** against any deployment records, infrastructure-as-code change history, or team communications.

Hypothesis D — Billing Anomaly or Data Artifact (UNVERIFIED) The observed cost reduction may not reflect a genuine infrastructure change but could instead represent a billing timing artifact, a credit application, a Reserved Instance or Savings Plan coverage change, or a data anomaly within the SYNTHETIC dataset itself. This has **not been confirmed** against any billing record, credit memo, or CUR data.

Summary No single hypothesis is preferred or ranked above others at this stage. All four scenarios are plausible given the available synthetic evidence. **Human verification is required before any hypothesis can be accepted, rejected, or acted upon.**

 This hypothesis has not been verified and must not be acted upon.

4.5 5. VERIFICATION REQUIRED

The following human verification steps are required to confirm or refute the hypothesis presented in Section 4. **No automated verification, no API calls, and no workload-changing actions may be performed as part of this verification process.**

All verification steps must be completed by a qualified human reviewer from the **Demo Platform Team** before any conclusion is drawn or any action is considered.

Step 1 — Review AWS CloudTrail Logs (Human Review Only) A human reviewer must manually inspect AWS CloudTrail logs for the ap-southeast-2 (Sydney) region covering the period **7–8 July 2026** to identify any EC2-related API events, including but not limited to:

- `StopInstances`
- `TerminateInstances`
- `ModifyInstanceAttribute` (instance type changes)
- `UpdateAutoScalingGroup`

This review must be performed by a human with appropriate IAM read permissions. No automated execution of this step is authorized by this report.

Step 2 — Review AWS Config Change History (Human Review Only) A human reviewer must inspect AWS Config configuration history for EC2 instances and Auto Scaling groups in ap-southeast-2 to identify any configuration state changes recorded between **1 July 2026 and 14 July 2026**. This will help establish whether instance counts, instance types, or scaling configurations changed between the baseline and measurement windows.

Step 3 — Review Instance Scheduler or Auto Scaling Policies (Human Review Only) A human reviewer must examine any active instance scheduler configurations, Auto Scaling scheduled actions, or operational runbooks applicable to the staging environment in ap-southeast-2 to determine whether any scheduled scaling or shutdown events were configured to take effect on or around **8 July 2026**.

Step 4 — Review Cost and Usage Report (CUR) Line Items (Human Review Only) A human reviewer with access to the actual AWS Cost and Usage Report (CUR) — noting that the current report is based on SYNTHETIC data only — must examine line-item detail for Amazon EC2 in ap-southeast-2 for the periods **1–7 July 2026** and **8–14 July 2026** to identify:

- Changes in instance hours billed
 - Changes in instance types or sizes
 - Any credits, refunds, or Reserved Instance/Savings Plan coverage changes applied
 - Any billing anomalies or data artifacts
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Step 5 — Consult the Demo Platform Team's Change Record (Human Review Only) A human reviewer must consult the Demo Platform Team's internal change management records, deployment logs, infrastructure-as-code (IaC) commit history, and team communications for the period **1–14 July 2026** to determine whether any planned or unplanned infrastructure changes were made to the staging environment that could account for the observed cost reduction.

Step 6 — Validate Measurement Window Comparability (Human Review Only) A human reviewer must confirm that the two measurement windows — **1–7 July 2026** (baseline) and **8–14 July 2026** (measurement) — are directly comparable in terms of:

- Number of calendar days included

- Business day vs. weekend composition
- Any public holidays in the ap-southeast-2 region that may affect workload patterns
- Consistency of workload type and volume across both windows

Step 7 — Document Verification Findings (Human Review Only) Upon completing Steps 1–6, the human reviewer must document their findings in writing, explicitly confirming or refuting each hypothesis presented in Section 4, and submit the documented findings to the appropriate FinOps governance process before any further action is considered.

 No automated or workload-changing verification may be performed. All steps above require human execution and human judgment.

4.6 6. OWNER

Assigned Owner: Demo Platform Team **Region of Responsibility:** ap-southeast-2 (Sydney) **Service in Scope:** Amazon EC2 — Staging Environment

Owner Responsibilities The **Demo Platform Team** is assigned as the responsible owner for this investigation and holds the following responsibilities in connection with this report:

Responsibility	Description
Primary Review	Conduct or delegate all human verification steps listed in Section 5 of this report.
Evidence Collection	Gather and document evidence from CloudTrail, AWS Config, CUR line items, and internal change records as specified in Section 5.
Hypothesis Assessment	Evaluate each hypothesis presented in Section 4 against the collected evidence and document a determination for each.
Escalation	Escalate findings to the appropriate FinOps governance body or engineering leadership if the root cause cannot be determined through standard verification steps.
Reporting	Provide a written verification outcome report to the FinOps function upon completion of the review, referencing this report by its classification and period (July 2026 [SYNTHETIC]).
Authorization Gate	No action — including any workload change, optimization implementation, or rollback — may proceed without explicit written authorization from the Demo Platform Team following completion of the verification process.
Timeline	The Demo Platform Team should establish and communicate a target timeline for completing the verification steps in Section 5, consistent with the materiality of the finding (SYNTHETIC USD 15.00 / 35.71% reduction).

The Demo Platform Team is the sole authorized body to progress this investigation beyond the read-only stage. This report does not delegate, transfer, or expand any authority beyond what is described above.

4.7 7. PROPOSED NEXT STEP: HUMAN REVIEW ONLY

 The only permitted next step arising from this report is human review by the Demo Platform Team.

This report is **read-only**. It has been produced for the purpose of informing and structuring a human-led investigation. The following constraints apply absolutely:

- **No automated action** has been initiated, is recommended, or is authorized by this report.

- **No workload-changing action** — including but not limited to instance modification, termination, scaling adjustment, policy change, or configuration update — has been executed, recommended, or approved.
- **No approval of execution** of any kind is granted by this report or by the workflow that produced it.
- **No external system** has been accessed, queried, or modified in the production of this report.

The Single Authorized Next Step The **Demo Platform Team** must:

1. Receive and review this report in full.
2. Assign a qualified human reviewer to execute the verification steps documented in Section 5.
3. Document verification findings in writing.
4. Determine, based on verified evidence, whether any further action is warranted.
5. Obtain appropriate authorization before any action is taken.

Until these steps are completed by a human reviewer, **this investigation remains open and no conclusions should be drawn from the unverified hypothesis in Section 4.**

4.8 8. RISK AND ROLLBACK

Risks of Acting Without Verification Acting on the unverified hypothesis in Section 4 — or on the observed cost reduction alone — without completing the human verification steps in Section 5 carries the following risks:

Risk	Description
Incorrect Root Cause Attribution	The cost reduction may have a cause entirely different from any hypothesis proposed. Acting on an incorrect assumption could result in unintended infrastructure changes or missed optimization opportunities.
Workload Disruption	If the cost reduction reflects an unintended or unauthorized infrastructure change (e.g., an accidental instance termination), acting to "preserve" the reduced cost state without verification could permanently remove required staging capacity.
Compliance and Governance Breach	Implementing workload changes based on unverified cost data may violate internal change management, FinOps governance, or audit requirements applicable to the Demo Platform Team.
Data Artifact Risk	If the observed reduction is a SYNTHETIC data artifact or billing anomaly rather than a genuine infrastructure change, any action taken in response would be based on false evidence.
Irreversibility	Certain infrastructure changes — particularly instance terminations or decommissions — may be difficult or impossible to reverse without significant effort, cost, or data loss.
Missed Escalation	If the cost reduction reflects a security incident, unauthorized access, or unplanned outage, acting to optimize rather than investigate could delay appropriate incident response.

Rollback Statement

 **No rollback plan has been activated, recommended, or authorized by this report.**

This report does not assess, design, approve, or initiate any rollback procedure. Any rollback of infrastructure changes — whether related to or independent of the cost movement observed in this report — requires:

1. **Human authorization** from the Demo Platform Team and/or appropriate engineering leadership.
2. **Verified evidence** that a rollback is warranted, based on the completed verification steps in Section 5.
3. **Compliance with the Demo Platform Team's standard change management process.**

No rollback may be initiated on the basis of this report alone.

4.9 9. ACTION AUTHORITY

 No workload-changing action has been executed, recommended, or approved by this workflow. This report is produced and provided under the following absolute constraints:

Constraint	Status
Workload-changing action executed	 None
Workload-changing action recommended	 None
Workload-changing action approved	 None
External system accessed	 None
Automated verification performed	 None
Rollback initiated or recommended	 None
Approval of execution granted	 None

This report is **read-only** and exists solely to present synthesized observations, a structured financial impact assessment, an unverified hypothesis, and a defined verification pathway for human review.

All action authority rests exclusively with the Demo Platform Team and any other human stakeholders designated under the applicable FinOps governance framework. This report confers no authority, grants no permissions, and initiates no processes beyond its own generation.

Any party receiving this report who believes an urgent workload or security issue may be present should follow their organization's standard incident response and escalation procedures independently of this report.

4.10 10. FINAL STATUS

 **REVIEW REQUIRED** — Awaiting human verification by the Demo Platform Team before any action may be considered.

Status Field	Value
Report Status	 REVIEW REQUIRED
Investigation Phase	Open — Unverified
Materiality Threshold Met	 Yes (SYNTHETIC 35.71% 20% AND SYNTHETIC USD 15.00 USD 10.00)
Hypothesis Status	 UNVERIFIED — Not Confirmed
Verification Completed	 No — Pending human review
Action Authorized	 No — No action authorized
Assigned Owner	Demo Platform Team
Data Classification	SYNTHETIC
Source Dashboard	CUDOS Dashboard Demo [Synthetic]
Analysis Period	July 2026
Region	ap-southeast-2 (Sydney)
Primary Service	Amazon EC2 (SYNTHETIC USD 693.30 net unblended cost)

This investigation will remain at  **REVIEW REQUIRED** status until the Demo Platform Team completes the human verification steps documented in Section 5, documents their findings, and formally updates the investigation status through the applicable FinOps governance process.

 **SYNTHETIC DEMONSTRATION** — Not Real CUR Billing Data

This report was generated using only the inputs provided. No external systems, APIs, or data sources were accessed. No workload-changing actions were executed, recommended, or approved. All cost figures are labeled SYNTHETIC and are

sourced exclusively from the CUDOS Dashboard Demo [Synthetic]. This report is read-only and is intended solely for human review by the Demo Platform Team.

End of Report — Cost Investigation Report / July 2026 [SYNTHETIC] / ap-southeast-2 (Sydney) / Demo Platform Team
